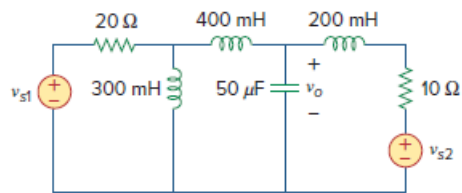


Problem

Use mesh analysis to find v_o in the circuit of Fig. 10.78. Let $v_{s1} = 120 \cos(100t + 90^\circ)$ V, $v_{s2} = 80 \cos 100t$ V.

Figure 10.78:



Step-by-step solution

Step 1 of 6

Consider the following circuit diagram:

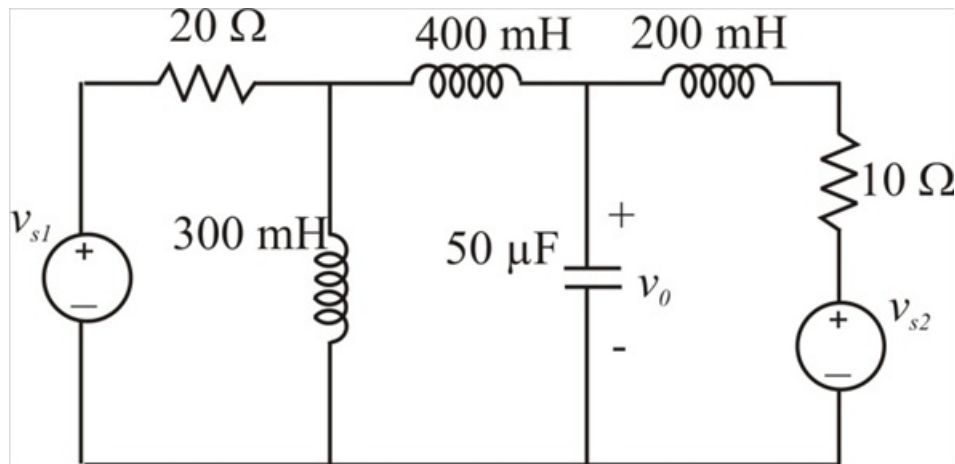


Figure 1

Step 2 of 6

Consider the following source voltages.

$$v_{s1} = 120 \cos(100t + 90^\circ) \text{ V}$$

$$v_{s2} = 80 \cos 100t \text{ V}$$

Convert the circuit to frequency domain.

$$\omega = 100 \text{ rad/s}$$

Convert the inductance value.

$$L = 300 \text{ mH}$$

$$\begin{aligned} j\omega L &= j(100 \text{ rad/s})(300 \text{ mH}) \\ &= j30 \Omega \end{aligned}$$

Convert the inductance value.

$$L = 200 \text{ mH}$$

$$\begin{aligned} j\omega L &= j(100 \text{ rad/s})(200 \text{ mH}) \\ &= j20 \Omega \end{aligned}$$

Convert the inductance value.

$$L = 400 \text{ mH}$$

$$\begin{aligned} j\omega L &= j(100 \text{ rad/s})(400 \text{ mH}) \\ &= j40 \Omega \end{aligned}$$

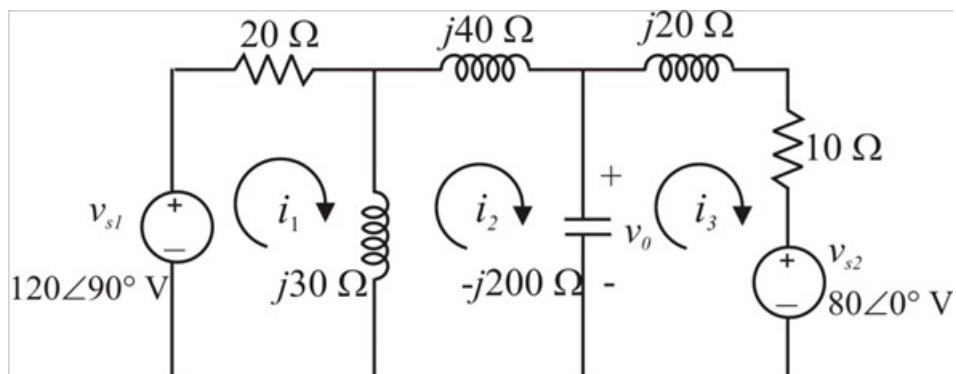
Convert the capacitance value.

$$C = 50 \mu\text{F}$$

$$\begin{aligned} \frac{1}{j\omega C} &= \frac{1}{j(100 \text{ rad/s})(50 \mu\text{F})} \\ &= \frac{1}{j0.005} \\ &= -j200 \Omega \end{aligned}$$

Step 3 of 6

Draw the equivalent frequency domain circuit:



Step 4 of 6

Figure 2

Step 5 of 6

Assume that the currents $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{I}_3$ are flowing in the circuit.

Apply Kirchhoff's voltage law to mesh 1.

$$\begin{aligned} 20\mathbf{I}_1 + j30(\mathbf{I}_1 - \mathbf{I}_2) &= 120\angle 90^\circ \\ (20 + j30)\mathbf{I}_1 - j30\mathbf{I}_2 &= j120 \\ (2 + j3)\mathbf{I}_1 - j3\mathbf{I}_2 &= j12 \quad \dots\dots (1) \end{aligned}$$

Apply Kirchhoff's voltage law to mesh 2.

$$\begin{aligned} j30(\mathbf{I}_2 - \mathbf{I}_1) + j40\mathbf{I}_2 - j200(\mathbf{I}_2 - \mathbf{I}_3) &= 0 \\ -j30\mathbf{I}_1 - j130\mathbf{I}_2 + j200\mathbf{I}_3 &= 0 \\ -j3\mathbf{I}_1 - j13\mathbf{I}_2 + j20\mathbf{I}_3 &= 0 \quad \dots\dots (2) \end{aligned}$$

Apply Kirchhoff's voltage law to mesh 3.

$$\begin{aligned} -j200(\mathbf{I}_3 - \mathbf{I}_2) + j20\mathbf{I}_3 + 10\mathbf{I}_3 &= -80\angle 0^\circ \\ j200\mathbf{I}_2 + (-j180 + 10)\mathbf{I}_3 &= -80 \\ j20\mathbf{I}_2 + (1 - j18)\mathbf{I}_3 &= -8 \quad \dots\dots (3) \end{aligned}$$

Represent the equations (1), (2) and (3) into matrix form.

$$\begin{bmatrix} 2 + j3 & -j3 & 0 \\ -3 & -13 & 20 \\ 0 & j20 & 1 - j18 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \\ \mathbf{I}_3 \end{bmatrix} = \begin{bmatrix} j12 \\ 0 \\ -8 \end{bmatrix}$$

Calculate the determinant values of matrices to find $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{I}_3$.

$$\begin{aligned} \Delta &= \begin{vmatrix} 2 + j3 & -j3 & 0 \\ -3 & -13 & 20 \\ 0 & j20 & 1 - j18 \end{vmatrix} \\ &= (2 + j3)(-13 - j166) + (j3)(-3 + j54) + 0 \\ &= 310 - j380 \end{aligned}$$

$$\begin{aligned} \Delta_2 &= \begin{vmatrix} 2 + j3 & j12 & 0 \\ -3 & 0 & 20 \\ 0 & -8 & 1 - j18 \end{vmatrix} \\ &= (2 + j3)(-20 \times -8) - j12(-3)(1 - j18) + 0 \\ &= 968 + j516 \end{aligned}$$

$$\begin{aligned} \Delta_3 &= \begin{vmatrix} 2 + j3 & -j3 & j12 \\ -3 & -13 & 0 \\ 0 & j20 & -8 \end{vmatrix} \\ &= (2 + j3)(-13 \times -8) + j3(-3 \times -8) + j12(-3 \times j20) \\ &= 928 + j384 \end{aligned}$$

[Comments \(1\)](#)



Anonymous

This solution setup is wrong: Pay close attention to the second row in the matrix representation of all three equations; that row is missing all the complex variables (j). Please fix

Calculate the current $\mathbf{I}_2$ in mesh 2:

$$\begin{aligned}\mathbf{I}_2 &= \frac{\Delta_2}{\Delta} \\ &= \frac{968 + j516}{310 - j380} \\ &= 0.4324 + j2.1945 \\ &= 2.236 \angle 78.85^\circ\end{aligned}$$

Calculate the current $\mathbf{I}_3$ in mesh 3:

$$\begin{aligned}\mathbf{I}_3 &= \frac{\Delta_3}{\Delta} \\ &= \frac{928 + j384}{310 - j380} \\ &= 0.589 + j1.961 \\ &= 2.047 \angle 73.27^\circ\end{aligned}$$

Calculate the value of voltage $\mathbf{V}_o$:

$$\begin{aligned}\mathbf{V}_o &= (\mathbf{I}_2 - \mathbf{I}_3)(-j200 \, \Omega) \\ &= (2.236 \angle 78.85^\circ - 2.047 \angle 73.27^\circ)(200 \angle -90^\circ) \\ &= 56.23 \angle 33.84^\circ\end{aligned}$$

Convert the voltage value $\mathbf{V}_o$ to time domain.

$$\mathbf{V}_o = 56.23 \cos(100t + 33.84^\circ) \, \text{V}$$

Thus, the value of voltage v_o is $\boxed{56.23 \cos(100t + 33.84^\circ) \, \text{V}}$.